

DBA Team Performance Tracking System

Executive Summary

This document describes the automated performance tracking system implemented for the DBA team at MTN Irancell. The system provides objective metrics for evaluating team productivity and attendance through Discord activity monitoring, email tracking, and automated reporting.

1. Background

The DBA team coordinates primarily through Discord. To address productivity monitoring needs and provide objective performance data, an automated tracking system was developed.

Objectives

- Track daily attendance (check-in/check-out times)
 - Monitor active work hours and break durations
 - Measure engagement through Discord activity
 - Track email communications
 - Generate automated reports for management review
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2. System Architecture

DBA Performance Tracking System

Discord Bot
(Real-time)

Discord Export
(Daily)

Email Extract
(Daily)

SQLite Database

attendance.db (idle, offline, voice, leave, remote_work)

Attendance Tracker

(attendance_tracker.py)

Daily CSV
Export

Email Report
(HTML)

Monthly
Report

3. File Structure

3.1 Core Files

File	Description
attendance_tracker.py	Main attendance analysis and reporting engine. Parses Discord messages, calculates work hours, generates CSV and SQLite exports
email_sender.py	Sends HTML email reports with CSV attachments via Exchange
monthly_report.py	Generates monthly aggregated reports in Persian (Jalali calendar)
attendance_db.py	SQLite database manager with CLI for queries
migrate_to_sqlite.py	One-time migration script from JSON to SQLite

3.2 Data Collection Files

File	Description
discord/discord_idle_bot.py	Real-time Discord status monitoring bot. Tracks idle/offline/voice time
discord/discord_exporter.py	Exports Discord channel messages to JSON files
email_extractor.py	Extracts email counts from Exchange (for attendance metrics)
email_full_extractor.py	Exports full email content (for mtn_update.py analysis)

3.3 Support Files

File	Description
build_leave_database.py	Parses historical Discord exports to build leave request database
remote_work_tracker.py	Parses and manages remote work records
leave_parser.py	Parses leave request messages from Discord
mtn_update.py	Analyzes received emails for work metrics
view_attendance.py	CLI tool to view attendance data

3.4 Automation Scripts

File	Description
daily_attendance_cron.sh	Main cron script that orchestrates daily reporting
monthly_attendance_cron.sh	Monthly report generation script
cleanup.sh	Cleanup temporary files

3.5 Data Storage

File/Table	Description
attendance_reports/attendance.db	Main SQLite database with all attendance data
attendance_reports/daily_attendance.csv	Historical CSV with all daily records
discord/activity.db	Discord bot session-level tracking
*.json	Legacy JSON files (kept as fallback)

4. Discord Bot (`discord_idle_bot.py`)

4.1 Purpose

Real-time monitoring of Discord presence status for all team members.

4.2 Tracked Statuses

Status	Description
Online	User is active on Discord
Idle	Discord shows user as idle (yellow dot)
Offline	User is not connected to Discord
Voice	User is in a voice channel

4.3 Features

- **Presence Tracking:** Monitors status changes in real-time
- **Voice Channel Tracking:** Records time spent in voice channels
- **Session Persistence:** Saves open sessions to survive bot restarts
- **SQLite Sync:** Syncs daily data to the unified SQLite database
- **Work Hours Only:** Only tracks during work hours (08:00-18:00)

4.4 Data Storage

```
# Data structures in memory
daily_idle_time = {date_str: {user_id: minutes}}
daily_offline_time = {date_str: {user_id: minutes}}
daily_voice_time = {date_str: {user_id: minutes}}

# Synced to SQLite tables
idle_tracking (date, name, idle_minutes)
offline_tracking (date, name, offline_minutes)
voice_tracking (date, name, voice_minutes)
```

4.5 Running the Bot

```
# Start the bot
cd /root/infrastructure/MTN_Repo/discord
source .env
python3 discord_idle_bot.py

# Or as a systemd service
sudo systemctl start discord-idle-tracker
```

5. Attendance Tracker (`attendance_tracker.py`)

5.1 Purpose

Parse Discord messages to extract attendance data and calculate work metrics.

5.2 Message Detection Patterns

Greetings (Check-in): - salam, sobh bekheir, rooz bekheir, hi, vorud

Farewells (Check-out): - khaste nabashid, felan, shab bekheir, bye, khodafez

Breaks (BRB): - brb - Start break - brb nahar - Lunch break (1 hour) - b, back - Return from break

Leave Requests: - morekhasi farda - Full day tomorrow - morekhasi az 14 ta 16 - Hourly leave - off, morakhasi with date ranges

5.3 Metrics Calculation

Gross Work Time = Check-out - Check-in

BRB Excess = BRB_total - Free_allowance (60 min)

Effective Hours = Gross - BRB - max(Idle, HourlyLeave) - Offline + Bonus

5.4 Command Line Usage

Generate today's report

```
python3 attendance_tracker.py
```

Export to CSV

```
python3 attendance_tracker.py --csv
```

Backfill historical data

```
python3 attendance_tracker.py --backfill --from 2025-04-21
```

Analyze specific date

```
python3 attendance_tracker.py --date 2025-12-28
```

6. Email Reports

6.1 Daily Report (email_sender.py)

Sent at 20:00 containing: - Attendance table for all team members - Check-in/check-out times - Work hours and BRB time - Idle/Offline percentages - Email and Discord message counts - Warnings for late arrival, early departure, low hours

Recipients: - Production: alireza.aghaja@, maryam.mare@, amirbahram.b@, hossein.mog@ - Test: alireza.aghaja@, maryam.mare@ only

6.2 Monthly Report (monthly_report.py)

Generated on 24th of each Jalali month containing: - Aggregated attendance data - Total work hours per person - Leave summary - Average daily metrics - Trend analysis

7. Database Schema

7.1 Main Attendance Table

```
CREATE TABLE attendance (  
    id INTEGER PRIMARY KEY,  
    date TEXT NOT NULL,  
    name TEXT NOT NULL,  
    check_in TEXT,  
    check_out TEXT,  
    work_hours REAL,  
    brb_minutes TEXT,
```

```

    emails INTEGER,
    discord INTEGER,
    voice INTEGER,
    effective_minutes REAL,
    leave TEXT,
    leave_hours TEXT,
    idle_minutes INTEGER,
    offline_minutes INTEGER,
    is_oncall TEXT,
    is_support TEXT,
    absent TEXT,
    UNIQUE(date, name)
);

```

7.2 Tracking Tables

```

CREATE TABLE idle_tracking (
    id INTEGER PRIMARY KEY,
    date TEXT NOT NULL,
    name TEXT NOT NULL,
    idle_minutes REAL,
    UNIQUE(date, name)
);

```

```

CREATE TABLE offline_tracking (
    id INTEGER PRIMARY KEY,
    date TEXT NOT NULL,
    name TEXT NOT NULL,
    offline_minutes REAL,
    UNIQUE(date, name)
);

```

```

CREATE TABLE voice_tracking (
    id INTEGER PRIMARY KEY,
    date TEXT NOT NULL,
    name TEXT NOT NULL,
    voice_minutes REAL,
    UNIQUE(date, name)
);

```

7.3 Leave and Remote Work

```

CREATE TABLE leave_records (
    id INTEGER PRIMARY KEY,
    date TEXT NOT NULL,
    name TEXT NOT NULL,
    leave_type TEXT,
    start_hour INTEGER,
    end_hour INTEGER,
    UNIQUE(date, name, leave_type, start_hour, end_hour)
);

```

```

CREATE TABLE remote_work (
    id INTEGER PRIMARY KEY,
    date TEXT NOT NULL,

```

```

    name TEXT NOT NULL,
    location TEXT,
    UNIQUE(date, name)
);

```

8. Working Hours Policy

8.1 Flexible Hours

Parameter	Value
Entry Window	08:00 - 09:30
Exit Window	17:00 - 18:30
Working Days	Saturday - Wednesday
Weekend	Thursday, Friday
Total Work Hours	9 hours (including 1 hour lunch)

8.2 Break Rules

Break Type	Allowance
Short Break (BRB)	30 min per session, 60 min total
Lunch Break (brb nahar)	60 minutes free

If a break exceeds the allowed duration, excess time is deducted from effective hours.

8.3 On-Call Rules

Day Type	Role	Effective Hours
Regular Day	On-call	9 hours automatic
Regular Day	Support (previous on-call)	9 hours automatic
Wednesday	On-call	Actual + 5 hour bonus

9. Automation Schedule (Cron)

Time	Task	Script
20:00 daily	Export Discord, generate report, send email	daily_attendance_cron.sh
10:00 on 24th	Generate monthly report	monthly_attendance_cron.sh

Crontab Entry:

```

0 20 * * * /root/infrastructure/scripts/daily_attendance_cron.sh >> /root/infrastructure/attendance_rep
0 10 * * * /root/infrastructure/scripts/monthly_attendance_cron.sh >> /root/infrastructure/attendance_r

```

10. Team Members

10.1 Active Members

Name	Discord Username	Status
Keivan Sadeghi	k1.sadeghi_15101	Active
Ehsan Yousefi	ehsan.yo	Active
Mohsen Roudsaz	mohsen.roud	Active
Nader Shabibi	nader3307	Active
Zeinabsadat Hejazi	mahsahejszi	Active
Hosseinali Shirali	hosseinshahreza	Active
Masoud Raffei	masoudraafiee	Active
Masoud Sereshki	masoudsereshki	Active
Yassin Alivand	nissay87	Active
Erfan Heidari	erfan_heidari	Active
Maryam Yousefi	maryam.you	Active

10.2 Excluded from Tracking

Name	Role	Reason
Alireza Aghajanzadeh	System Owner	Excluded
Maryam Marefati	Team Lead	Excluded
Hossein Feizollahi	Senior DBA	Excluded

11. Setup and Installation

11.1 Requirements

```
pip install discord.py exchangelib jdatetime pandas
```

11.2 Environment Variables

```
# Discord Bot
DISCORD_BOT_TOKEN=your_bot_token

# Email (Exchange)
SMTP_PASSWORD=your_exchange_password
```

11.3 Initial Setup

```
# Clone repository
git clone github.com/alireza8580/MTN_Repo

# Create symlink for backward compatibility
ln -s /root/infrastructure/MTN_Repo /root/infrastructure/scripts

# Install systemd service for Discord bot
sudo cp discord/discord-idle-tracker.service /etc/systemd/system/
sudo systemctl enable discord-idle-tracker
sudo systemctl start discord-idle-tracker
```

11.4 Migration from JSON to SQLite

```
# Run migration (one-time)
python3 migrate_to_sqlite.py
```

12. Troubleshooting

12.1 Common Issues

Issue	Solution
Bot shows offline users as idle	Check Discord presence intent permissions
Email not sending	Verify SMTP_PASSWORD environment variable
Missing attendance data	Check Discord export files in discord_exports/
SQLite errors	Ensure attendance.db has correct permissions

12.2 Log Locations

Log	Location
Daily cron	/root/infrastructure/attendance_reports/cron.log
Monthly cron	/root/infrastructure/attendance_reports/monthly_cron.log
Discord bot	stdout when running in terminal

13. Future Enhancements

- Integration with ticketing system (iCare) for task tracking
 - Outlook calendar integration for meeting tracking
 - Dashboard for real-time monitoring
 - Expansion to other teams
 - Machine learning for anomaly detection
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